

IP Protocol Numbers and Traffic Types

At the end of this episode, I will be able to:

1. Identify and explain the purpose of IP protocol numbers and traffic types, given a scenario.

Learner Objective: *Identify and explain the purpose of IP protocol numbers and traffic types, given a scenario.*

Description: In this episode, the learner will examine IP protocol numbers used in IP headers to identify upper level protocols. We will also explore traffic types such as unicast, broadcast and more.

- Internet Protocol (IP) types
 - o Internet Control Message Protocol (ICMP)
 - ♠ IP Protocol ID - 1
 - o Internet Group Management Protocol (IGMP)
 - ♠ IP Protocol ID - 2
 - o Transmission Control Protocol (TCP)
 - ♠ IP Protocol ID - 6
 - o User Datagram Protocol (UDP)
 - ♠ IP Protocol ID - 17
 - o Generic Routing Encapsulation (GRE)
 - ♠ IP Protocol ID -
 - o Internet Protocol Security (IPSec)
 - ♠ Authentication Header (AH)
 - ♠ IP Protocol ID - 51
 - ♠ Encapsulating Security Payload (ESP)
 - ♠ IP Protocol ID - 50
 - Traffic types
 - o Unicast - one-to-one communication
 - o Multicast - one-to-many communication
 - o Broadcast - one-to-all communication
 - o Anycast - one-to-the-best communication
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- Additional Resources:
 - o IANA IP Protocol Numbers - <https://www.iana.org/assignments/protocol-numbers/protocol-numbers.xml>